

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year:2025-2026
CourseCode	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/II	Regulation	R23
Date and Day of Assignment	Week4 – Monday	Time(s)	23CSBTB01 To 23CSBTB52
Name	Aelishala Manoj	Hall ticket No	2303A54054
Assignment Number: 8.1			
Q.No.	Question		
1	Lab 8: Test-Driven Development with AI – Generating and Working with Test Cases Lab Objectives: - To introduce students to test-driven development (TDD) using AI code generation tools. - To enable the generation of test cases before writing code implementations. - To reinforce the importance of testing, validation, and error handling. - To encourage writing clean and reliable code based on AI-generated test expectations. Lab Outcomes (LOs): After completing this lab, students will be able to: - Use AI tools to write test cases for Python functions and classes. - Implement functions based on test cases in a test-first development style. - Use unittest or pytest to validate code correctness. - Analyze the completeness and coverage of AI-generated tests. - Compare AI-generated and manually written test cases for quality and logic Task Description #1 (Password Strength Validator – Apply AI in Security Context) - Task: Apply AI to generate at least 3 assert test cases for <code>is_strong_password(password)</code> and implement the validator function. - Requirements: - Password must have at least 8 characters. - Must include uppercase, lowercase, digit, and special character. - Must not contain spaces. Example Assert Test Cases: <pre>assert is_strong_password("Abcd@123") == True assert is_strong_password("abcd123") == False assert is_strong_password("ABCD@1234") == True</pre> Expected Output #1: - Password validation logic passing all AI-generated test cases.		

```
lab8.1-4054.py X
lab8.1-4054.py > is_strong_password
4 def is_strong_password(password):
10     - Has at least one lowercase letter
11     - Has at least one digit
12     - Has at least one special character (non-alphanumeric)
13     """
14     if len(password) < 8 or " " in password:
15         return False
16
17     # Check for required character types using regex
18     # The regex checks for at least one occurrence of:
19     # [A-Z] : Uppercase
20     # [a-z] : Lowercase
21     # [0-9] : Digit
22     # [^\w] : Special character (anything not a word character)
23     # Note: \w includes [a-zA-Z0-9_], so underscore is considered a word character here.
24     patterns = [r"[A-Z]", r"[a-z]", r"\d", r"[^\w]"]
25
26     return all(re.search(p, password) for p in patterns)
27
28 # Test cases
29 assert is_strong_password("Abcd@123") == True
30 assert is_strong_password("abcd123") == False
31 assert is_strong_password("ABcd1234") == False # no special char
32 assert is_strong_password("ABCD@1234") == False # no lowercase
33 assert is_strong_password("Abc d@123") == False # contains space
34 print("Task 1: All tests passed.")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE DBCODE
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & C:/Users/alish/AppData/Local/Microsoft/WindowsApps/python3.12.exe c:/Users/a
lish/OneDrive/Documents/Desktop/AI-AS/lab8.1-4054.py
Task 1: Function defined successfully.
True
False
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & C:/Users/alish/AppData/Local/Microsoft/WindowsApps/python3.12.exe c:/Users/a
lish/OneDrive/Documents/Desktop/AI-AS/lab8.1-4054.py
Task 1: All tests passed.
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS>
```

Task Description #2 (Number Classification with Loops – Apply AI for Edge Case Handling)

- Task: Use AI to generate at least 3 assert test cases for a `classify_number(n)` function. Implement using loops.
- Requirements:
 - Classify numbers as Positive, Negative, or Zero.
 - Handle invalid inputs like strings and None.
 - Include boundary conditions (-1, 0, 1).

Example Assert Test Cases:

```
assert classify_number(10) == "Positive"
```

```
assert classify_number(-5) == "Negative"
```

```
assert classify_number(0) == "Zero"
```

Expected Output #2:

- Classification logic passing all assert tests.

```
lab8.1-4054.py
C:\Users\alish\OneDrive\Documents\Desktop\AI-AS\lab8.1-4054.py
36 #Task2
37 def classify_number(n):
38     """
39     Classifies a number as Positive, Negative, or Zero.
40     Returns 'Invalid Input' for non-numeric types.
41     """
42     if not isinstance(n, (int, float)):
43         return "Invalid Input"
44
45     if n > 0:
46         return "Positive"
47     elif n < 0:
48         return "Negative"
49     else:
50         return "Zero"
51
52 # Test cases
53 assert classify_number(10) == "Positive"
54 assert classify_number(-5) == "Negative"
55 assert classify_number(0) == "Zero"
56 assert classify_number(1) == "Positive"
57 assert classify_number(-1) == "Negative"
58 assert classify_number("abc") == "Invalid Input"
59 assert classify_number(None) == "Invalid Input"
60 print("Task 2: All tests passed.")
61
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE DBCODE
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & c:/Users/alish/AppData/Local/Microsoft/WindowsApps/python3.12.exe c:/Users/alish/OneDrive/Documents/Desktop/AI-AS/lab8.1-4054.py
Task 1: All tests passed.
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS>
Ln 59, Col 48 Spaces: 4 UTF-8 CRLF {} Python 3.12.10 (Microsoft Store)
```

Task Description #3 (Anagram Checker – Apply AI for String Analysis)

- Task: Use AI to generate at least 3 assert test cases for `is_anagram(str1, str2)` and implement the function.
- Requirements:
 - Ignore case, spaces, and punctuation.
 - Handle edge cases (empty strings, identical words).

Example Assert Test Cases:

```
assert is_anagram("listen", "silent") == True
```

```
assert is_anagram("hello", "world") == False
```

```
assert is_anagram("Dormitory", "Dirty Room") == True
```

Expected Output #3:

- Function correctly identifying anagrams and passing all AI-generated tests.

```
lab8.1-4054.py
lab8.1-4054.py > ...
62
63 #Task3
64 from collections import Counter
65
66 def is_anagram(str1, str2):
67     """
68     Checks if two strings are anagrams, ignoring case, spaces, and punctuation.
69     Uses Counter for O(n) complexity.
70     """
71     def clean_counter(s):
72         # Generator expression to filter and normalize characters
73         return Counter(ch.lower() for ch in s if ch.isalnum())
74
75     return clean_counter(str1) == clean_counter(str2)
76
77 # Test case
78 assert is_anagram("listen", "silent") == True
79 assert is_anagram("hello", "world") == False
80 assert is_anagram("Dormitory", "Dirty Room") == True
81 assert is_anagram("", "") == True # empty strings
82 assert is_anagram("Tea", "Eat") == True # case-insensitive
83 assert is_anagram("A gentleman!", "Elegant man") == True # punctuation ignored
84
85 print("Task 3: All tests passed.")
86
87
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE DBCODE
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & C:/Users/alish/AppData/Local/Microsoft/WindowsApps/python3.12.exe c:/Users/a
lsh/OneDrive/Documents/Desktop/AI-AS/lab8.1-4054.py
Task 1: All tests passed.
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS>
Ln 77, Col 13 Spaces: 4 UTF-8 CRLF {} Python 3.12.10 (Microsoft Store)
```

Task Description #4 (Inventory Class – Apply AI to Simulate Real-World Inventory System)

- Task: Ask AI to generate at least 3 assert-based tests for an Inventory class with stock management.
- Methods:
 - add_item(name, quantity)
 - remove_item(name, quantity)
 - get_stock(name)

Example Assert Test Cases:

```
inv = Inventory()
inv.add_item("Pen", 10)
assert inv.get_stock("Pen") == 10
inv.remove_item("Pen", 5)
assert inv.get_stock("Pen") == 5
inv.add_item("Book", 3)
assert inv.get_stock("Book") == 3
```

Expected Output #4:

- Fully functional class passing all assertions.

```
lab8.1-4054.py
lab8.1-4054.py > ...
89 class Inventory:
90     def add_item(self, name, quantity):
91         self.items[name] = self.items.get(name, 0) + quantity
92
93     def remove_item(self, name, quantity):
94         """Removes quantity from an item. Ensures stock doesn't go below zero."""
95         if name in self.items and quantity > 0:
96             current_stock = self.items[name]
97             # Use max to ensure stock doesn't drop below zero
98             self.items[name] = max(0, current_stock - quantity)
99
100     def get_stock(self, name):
101         """Returns the current stock of an item, defaulting to 0."""
102         return self.items.get(name, 0)
103
104 inv = Inventory()
105
106 # Test cases
107 inv.add_item("Pen", 10)
108 assert inv.get_stock("Pen") == 10
109
110 inv.remove_item("Pen", 5)
111 assert inv.get_stock("Pen") == 5
112
113 inv.add_item("Book", 3)
114 assert inv.get_stock("Book") == 3
115
116 inv.remove_item("Pen", 10) # removing more than available
117 assert inv.get_stock("Pen") == 0
118
119 assert inv.get_stock("Pencil") == 0 # item not added yet
120
121 print("Task 4: All tests passed.")
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```

Task Description #5 (Date Validation & Formatting – Apply AI for Data Validation)

- Task: Use AI to generate at least 3 assert test cases for `validate_and_format_date(date_str)` to check and convert dates.
- Requirements:
 - Validate "MM/DD/YYYY" format.
 - Handle invalid dates.
 - Convert valid dates to "YYYY-MM-DD".

Example Assert Test Cases:

```
assert validate_and_format_date("10/15/2023") == "2023-10-15"
```

```
assert validate_and_format_date("02/30/2023") == "Invalid Date"
```

```
assert validate_and_format_date("01/01/2024") == "2024-01-01"
```

Expected Output #5:

- Function passes all AI-generated assertions and handles edge cases.

```
lab8.1-4054.py
lab8.1-4054.py > ...
89 class Inventory:
93     def add_item(self, name, quantity):
96         self.items[name] = self.items.get(name, 0) + quantity
97
98     def remove_item(self, name, quantity):
99         """Removes quantity from an item. Ensures stock doesn't go below zero."""
100         if name in self.items and quantity > 0:
101             current_stock = self.items[name]
102             # Use max to ensure stock doesn't drop below zero
103             self.items[name] = max(0, current_stock - quantity)
104
105     def get_stock(self, name):
106         """Returns the current stock of an item, defaulting to 0."""
107         return self.items.get(name, 0)
108
109 inv = Inventory()
110
111 # Test cases
112 inv.add_item("Pen", 10)
113 assert inv.get_stock("Pen") == 10
114
115 inv.remove_item("Pen", 5)
116 assert inv.get_stock("Pen") == 5
117
118 inv.add_item("Book", 3)
119 assert inv.get_stock("Book") == 3
120
121 inv.remove_item("Pen", 10) # removing more than available
122 assert inv.get_stock("Pen") == 0
123
124 assert inv.get_stock("Pencil") == 0 # item not added yet
125
126 print("Task 4: All tests passed.")
127
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS QUERY RESULTS AZURE DECODE

PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS> & C:/Users/alish/AppData/Local/Microsoft/WindowsApps/python3.12.exe c:/Users/a
ish/OneDrive/Documents/Desktop/AI-AS/lab8.1-4054.py
Task 1: All tests passed.
PS C:\Users\alish\OneDrive\Documents\Desktop\AI-AS>

Ln 119, Col 34 Spaces: 4 UTF-8 CRLF Python 3.12.10 (Microsoft Store)